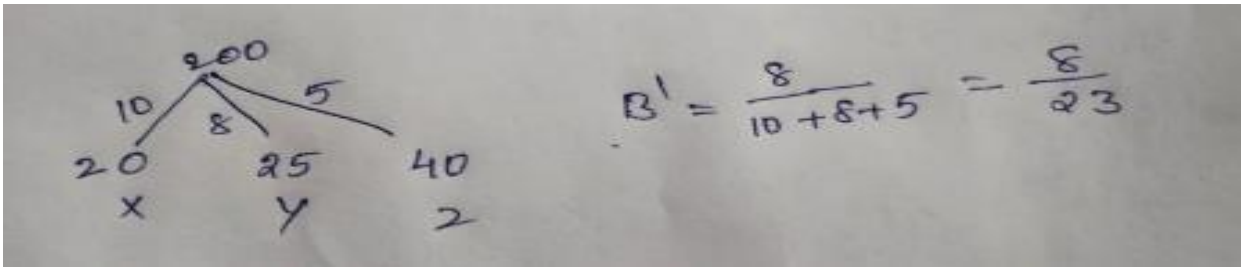
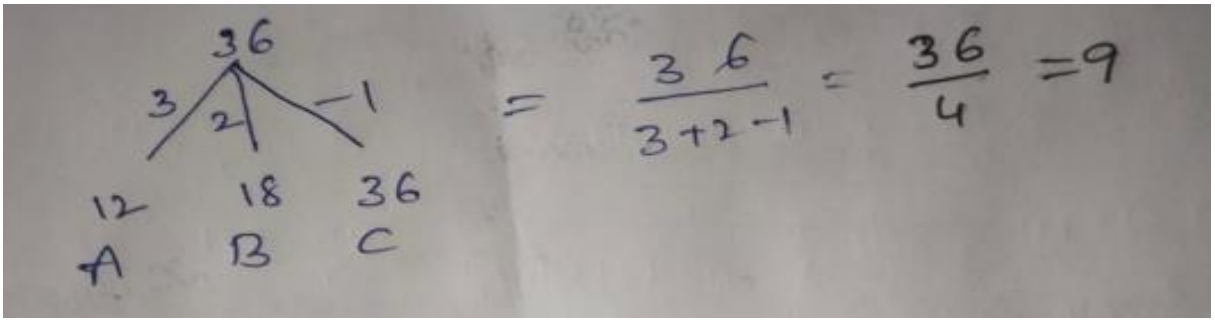
MODULE I : (TIME & WORK)

Short Answers

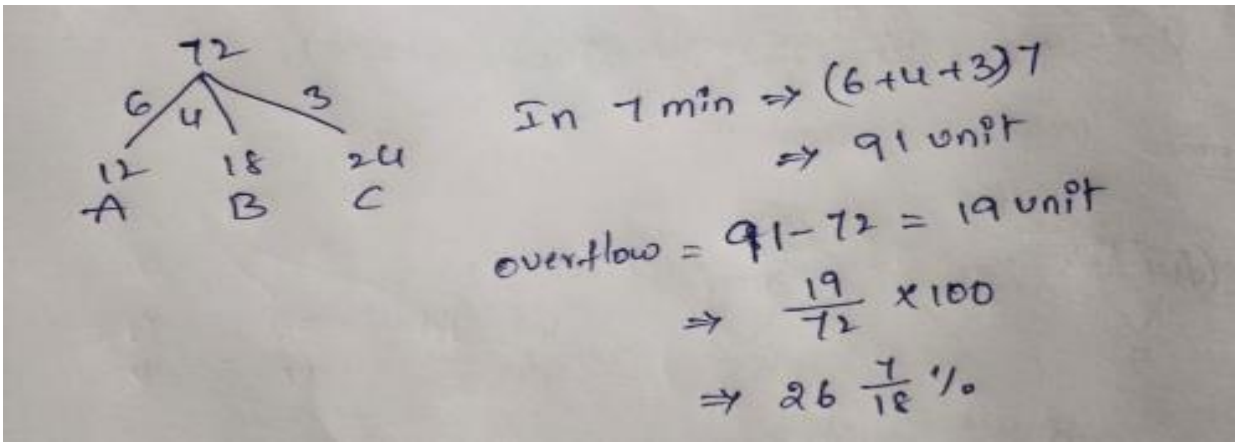
1. Three pipes X, Y and Z release three different chemicals A, B and C in a tank. These pipes can fill the tank in 20, 25 and 40 minutes respectively. If all the pipes are left open for 10 minutes, what will be the ratio of chemical B in the tank?



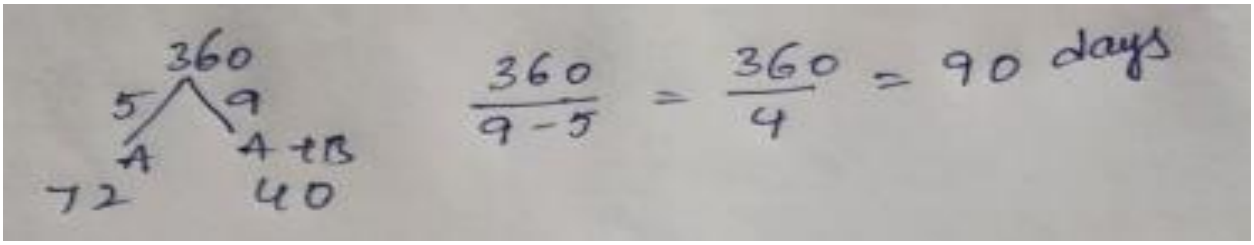
2. Pipe A can fill a tank in 12 minutes, pipe B can fill it in 18 minutes, while pipe C can empty the full tank in 36 minutes. If all the pipes are opened simultaneously, how much time will it take to fill the empty tank completely?



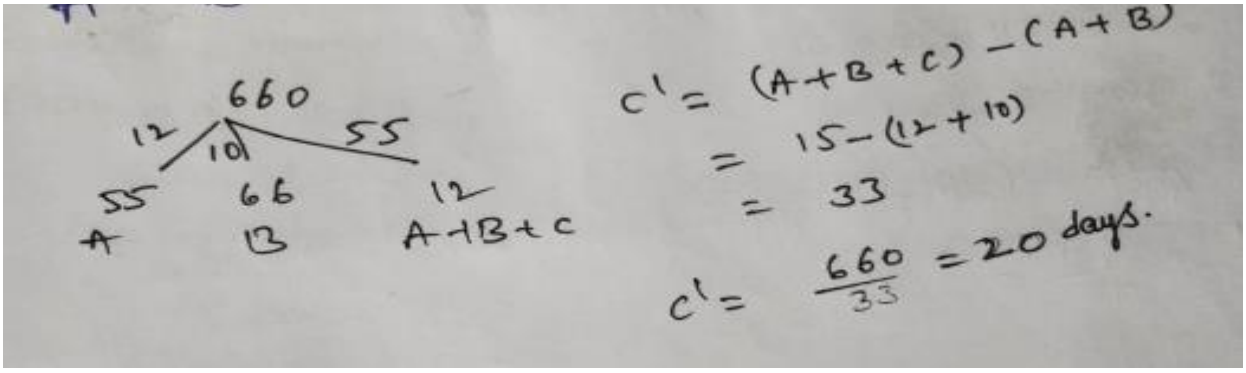
3. Three pipes A, B and C can fill a cistern in 12, 18 and 24 minutes, respectively. If all the pipes are opened together for 7 minutes, what will be the volume of the water that overflows as the percentage of the total volume of the cistern?



4. A alone can do a job in 72 days then, in how many days can B alone do the job if together they can do the job in 40 days?



5. A can paint a house in 55 days and B can do it in 66 days. Along with C, they did the job in 12 days only. Then, C alone can do the job in how many days?



6. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

Take LCM of days

LCM of 20, 30, 60 = 60
So total work = 60 units

Step 2: Daily work

- **A = $60/20 = 3$units/day**
- **B = $60/30 = 2$units/day**
- **C = $60/60 = 1$unit/day**

Step 3: Work pattern (3 days)

Day 1: A → 3 units
Day 2: A → 3 units
Day 3: A + B + C → 3 + 2 + 1 = 6 units

Work in 3 days = 3 + 3 + 6 = 12 units

Step 4: Total work

Total work = 60 units

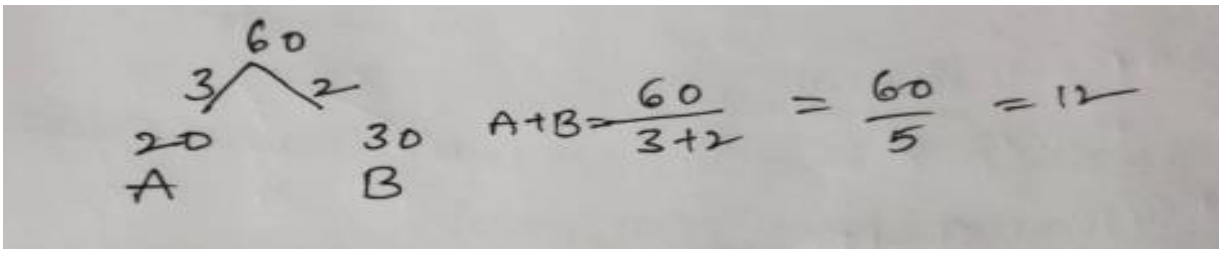
$$60 \div 12 = 5$$

Each cycle = 3 days

$$5 \times 3 = 15 \text{ days}$$

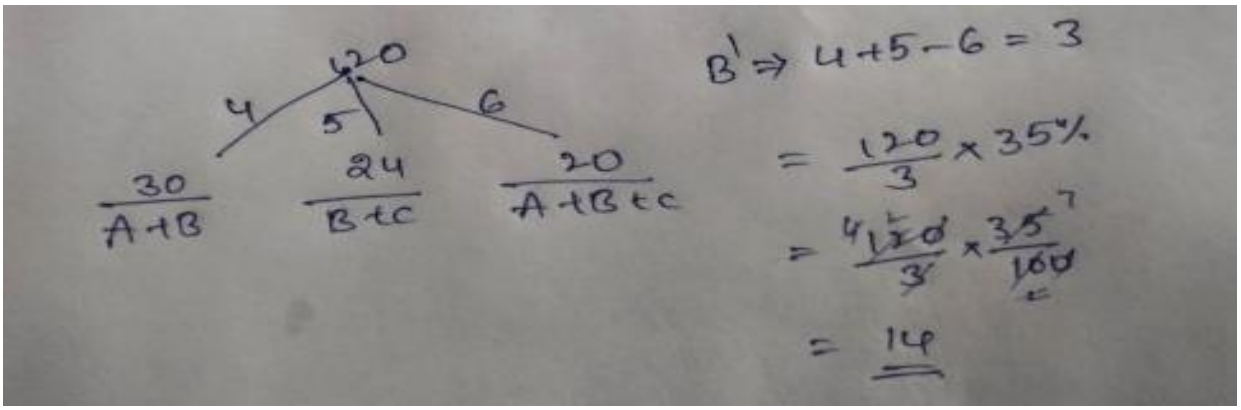
Answer: 15 days

7. Two pipes A and B can fill a tank in 20 and 30 minutes respectively. If both the pipes are used together, then how long will it take to fill the them?



Long Answers

1. Pipes A and B together can fill an empty tank in 30 hours, whereas pipes B and C together can fill it in 24 hours. A. B and C together can fill the tank in 20 hours. In how many hours can pipe B alone fill 35% of the tank?



2. Pipes A and B can fill a tank in 15 hours and 12 hours, respectively. Pipe C alone can empty the full tank in 10 hours. If all the three pipes are opened together for 2 hours 40 minutes, then what part of the tank will remain unfilled?

60

4 5 −6

15 12 10

A B C

2 Hr 40min = $2 + \frac{2}{3} = \frac{8}{3}$ Hrs

(A+B+C) in $\frac{8}{3}$ Hrs = $(4+5-6)\frac{8}{3}$

$= \cancel{3} \times \frac{8}{\cancel{3}}$

$= 8$

fraction of tank unfilled = $\frac{60-8}{60}$

$= \frac{52}{60} = \frac{13}{15}$.

3. A tank is normally filled in 20 hours by a pipe, but it takes 8 hours more to fill the tank due to a leakage at its bottom. The leakage point can empty the tank when it is 60% full in hours.

140

20 28

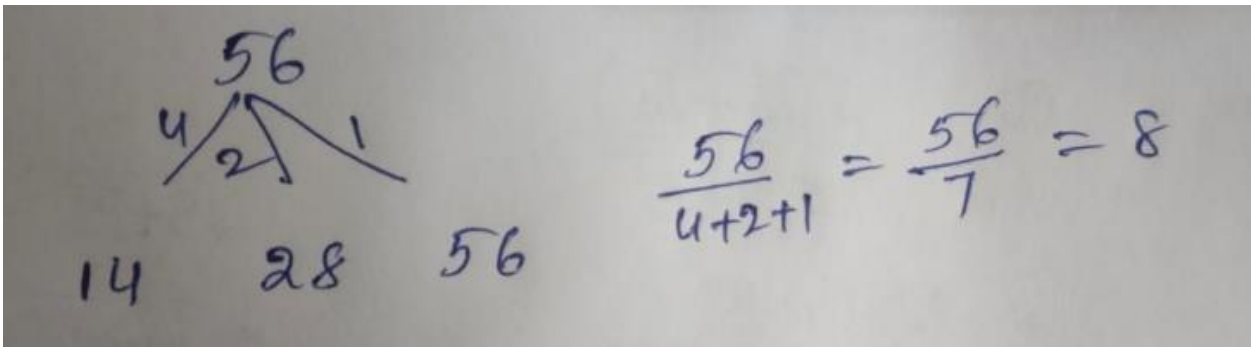
I I+O

⇒ eff of Leakage = 7−5 = 2

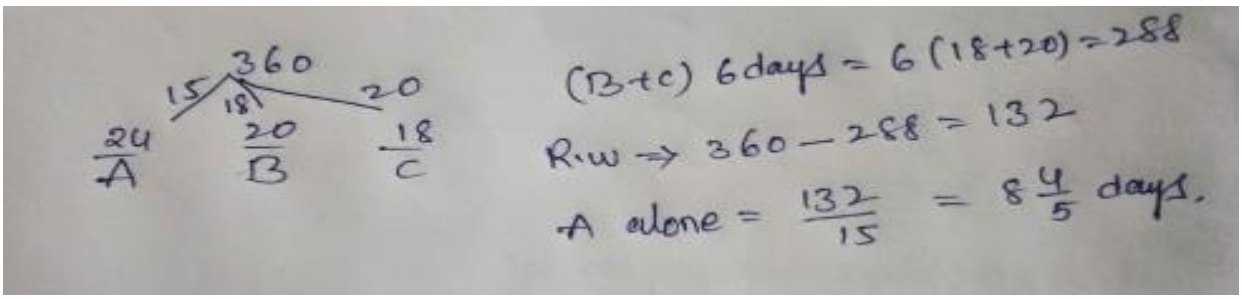
⇒ $\frac{140}{2} \times 60\%$

⇒ $\frac{140}{\cancel{2}} \times \frac{\overset{3}{\cancel{60}}}{\cancel{100}} = 42$ Hrs.

4. A tyre has 3 punctures. The first puncture alone would have made the tyre flat in 14 minutes, the second alone would have done it in 28 minutes, the third alone would have done it in 56 minutes. If air leaks out at a constant rate, how long (in minutes) does it take for all the punctures together to make the tyre flat?



5. A, B, and C can individually complete a task in 24 days, 20 days and 18 days, respectively B and C start the task, and they work for 6 days and leave. The number of days required by A alone to finish the remaining task?



6.A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?

Individual work rates

- A can complete the work in 6 days → A's 1-day work = $\frac{1}{6}$
- B can complete the work in 8 days → B's 1-day work = $\frac{1}{8}$

A and B working together

$$\frac{1}{6}+\frac{1}{8}$$

LCM of 6 and 8 = 24

$$= \frac{4}{24} + \frac{3}{24} = \frac{7}{24}$$

So, A and B together can complete 7/24 of the work in one day.

Work done in 3 days (A + B)

$$3 \times \frac{7}{24} = \frac{21}{24} = \frac{7}{8}$$

So A and B together complete 7/8 of the work in 3 days.

Work done by C in 3 days

Total work = 1

Remaining work: $1 - \frac{7}{8} = \frac{1}{8}$

So C completes 1/8 of the work in 3 days.

C's 1-day work:

$$\frac{1}{8} \div 3 = \frac{1}{24}$$

Payment Distribution

Total payment = Rs. 3200

Work ratio:

- A's 1-day work = 1/6 = 4/24
- B's 1-day work = 1/8 = 3/24
- C's 1-day work = 1/24

Ratio: $A:B:C = 4:3:1$
Total parts = 8

C's share: $\frac{1}{8} \times 3200 = 400$
Amount to be paid to C = Rs. 400

7. A and B together can do a piece of work in 30 days. A having worked for 16 days, B finishes the remaining work alone in 44 days. In how many days shall B finish the whole work alone?

A and B together complete the work in 30 days.

So, in 1 day they complete $\frac{1}{30}$of the work.

Work done in first 16 days by A and B $\quad 16 \times \frac{1}{30} = \frac{16}{30} = \frac{8}{15}$

Remaining work: $\quad 1 - \frac{8}{15} = \frac{7}{15}$

Remaining work done by B alone in 44 days

$$44B = \frac{7}{15}$$

So B's 1-day work:

$$B = \frac{7}{15 \times 44} = \frac{7}{660}$$

Time taken by B alone to complete the whole work

$$\text{Time} = \frac{1}{B} = \frac{660}{7}$$
$$= 94\frac{2}{7} \text{ days}$$

B alone can finish the whole work in $\frac{660}{7}$days ≈ 94.29 days.

8. A tank is filled by three pipes with uniform flow. The first two pipes operating simultaneously fill the tank in the same time during which the tank is filled by the third pipe alone. The second pipe fills the tank 5 hours faster than the first pipe and 4 hours slower than the third pipe. The time required by the first pipe is?

Let the time taken by the first pipe = $x$hours.

Then according to the question:

Second pipe fills 5 hours faster than the first → $x - 5$

Second pipe fills 4 hours slower than the third → Third pipe time = $x - 9$

So the times are:

First pipe = x

Second pipe = $x - 5$

Third pipe = $x - 9$

Use the given condition

The first two pipes together take the same time as the third pipe alone.

So their rates are equal:

Solve the equation

$$\frac{x - 5 + x}{x(x-5)} = \frac{1}{x-9}$$
$$\frac{2x - 5}{x(x-5)} = \frac{1}{x-9}$$

Cross multiply:

$$(2x-5)(x-9) = x(x-5)$$

Expand:

$$2x^2 - 23x + 45 = x^2 - 5x$$
$$x^2 - 18x + 45 = 0$$

Solve:

$$x = \frac{18 \pm \sqrt{324 - 180}}{2}$$
$$x = \frac{18 \pm 12}{2}$$
$$x = 15 \text{ or } 3$$

Since time cannot be less than 9 hours (because third pipe $x - 9$must be positive),

$$x = 15$$

The first pipe takes 15 hours to fill the tank.

MODULE II: (PnC & PROBABILITY)

Short Answers

1. In how many ways can a committee consisting of 4 men and 2 women can be chosen from 6 men and 5 women?

Choose 4 men from 6

$$^{6}C_4 = \frac{6!}{4!\,2!}$$
$$= \frac{6 \times 5}{2 \times 1} = 15$$

Choose 2 women from 5

$$^{5}C_2 = \frac{5!}{2!\,3!}$$
$$= \frac{5 \times 4}{2 \times 1} = 10$$

Total committees

$15 \times 10 = 150$

Total number of committees = 150.

2. A committee of 5 members is to be formed from a group of 10 people. In how many ways can this be done?

We need to choose 5 members from 10 people.

Step 1: Use combination formula

Here
$n = 10, r = 5$

$$^{10}C_5 = \frac{10!}{5!\,5!}$$

Simplify

$$^{10}C_5 = \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1}$$
$$= 252$$

Number of ways = 252.

3. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red?

First find the total number of balls.

White = 4 Red = 5 Blue = 6

Total balls = $4 + 5 + 6 = 15$

Total ways to draw 3 balls from 15

$$^{15}\boldsymbol{C_3}$$

Ways to draw 3 red balls from 5 red balls

$$^{5}C_{3}$$

Probability formula

P(E)={Favorable outcomes}/{Total outcomes}}

$$P(\text{all red}) = \frac{^{5}C_{3}}{^{15}C_{3}}$$

Calculate

$$^{5}C_{3} = 10$$

$$^{15}C_{3} = \frac{15 \times 14 \times 13}{6} = 455$$

$$P = \frac{10}{455} = \frac{2}{91}$$

Probability that all three balls are red = $\frac{2}{91}$.

4. In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?

Total lottery tickets

Prizes = 10 Blanks = 25

Total tickets = $10 + 25 = 35$

Use probability formula

P(E)={Number of favorable outcomes}/{Total number of outcomes}

Substitute values

Favorable outcomes (getting a prize) = 10
Total outcomes = 35

$P(\text{prize}) = \frac{10}{35} = \frac{2}{7}$

Probability of getting a prize = $\frac{2}{7}$.

5. A coin is thrown 3 times. what is the probability that at least one head is obtained?

Each toss has 2 outcomes (H or T).

Total outcomes for 3 tosses

$$2^3 = 8$$

Possible outcomes:
HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Event – *At least one head*

"At least one head" means 1 head, 2 heads, or 3 heads.

Instead of counting all these, use the complement method.

Event complement: No head (all tails)
Outcome: TTT

Probability of all tails:

P(TTT) = $P(TTT) = \frac{1}{8}$

Required probability

$$P(\text{at least one head}) = 1 - P(\text{no head})$$

$$= 1 - \frac{1}{8} = \frac{7}{8}$$

Final Answer: $\boxed{\frac{7}{8}}$

6. What is the probability of getting a sum of 7 when two dice are thrown?

Each die has 6 outcomes, so:

$$6 \times 6 = 36$$

Total outcomes = 36

Outcomes giving sum = 7

Possible pairs:

(1,6) (2,5) (3,4) (4,3) (5,2) (6,1)

Favorable outcomes = 6

S Probability formula

P(E) = {Number of favorable outcomes}/ {Total number of outcomes}

$$P(\text{sum } = 7) = \frac{6}{36} = \frac{1}{6}$$

7. From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

From 7 men and 6 women, a committee of 5 persons is to be formed with at least 3 men.

Possible cases "At least 3 men" means:

3 men and 2 women, 4 men and 1 woman, 5 men and 0 women

3 men and 2 women

Select 3 men from 7 and 2 women from 6

$$^{7}C_3 \times^{6} C_2$$
$$= 35 \times 15 = 525$$

4 men and 1 woman

$$^{7}C_4 \times^{6} C_1$$
$$= 35 \times 6 = 210$$

5 men and 0 women

$$^{7}C_5 \times^{6} C_0$$
$$= 21 \times 1 = 21$$

Total ways

$$525 + 210 + 21 = 756$$

8. In how many different ways can the letters of the word 'CHAMPIONS' be arranged in such a way that the vowels always come together?

Vowels: A, I, O → 3
Consonants: C, H, M, P, N, S → 6

Total letters = 9

Treat vowels as one block

Since vowels must stay together, consider them as one group.

So, items to arrange =

1 vowel block 6 consonants

Total = 7 items

Arrange these 7 items

$$7! = 5040$$

Arrange vowels inside the block

Vowels A, I, O are all different.

$$3! = 6$$

Total arrangements

$$7! \times 3!$$
$$= 5040 \times 6$$
$$= 30240$$

Long Answers

1. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

Here, S = {1, 2, 3, 4, ..., 19, 20}.

Let E = event of getting a multiple of 3 or 5 = {3, 6 , 9, 12, 15, 18, 5, 10, 20}.

P(E) = n(E)/n(S) = 9/20.

2. A bag contains 6 white and 4 black balls .2 balls are drawn at random. Find the probability that they are of same colour.

Let S be the sample space

Then n(S) = no of ways of drawing 2 balls out of (6+4) =$10C_2$ 10 = $\frac{10*9}{2*1}$ =45

Let E = event of getting both balls of same colour

Then,n(E) = no of ways (2 balls out of six) or (2 balls out of 4)

$$=6C_2 + 4C_2 = \frac{6*5}{2*1} + \frac{4*3}{2*1} = 15+6 = 21$$

Therefore, P(E) = n(E)/n(S) = 21/45 = 7/15

3. one card is drawn at random from the pack of 52 cards.

a) Find the Probability that it is an honor card.

Honor cards are: A, K, Q, J

Each suit has 4 honor cards, and there are 4 suits.

So total honor cards =

$$4 \times 4 = 16$$

Using probability formula:

P(E)= {Favorable outcomes}/ {Total outcomes}

$$P(\text{Honor card}) = \frac{16}{52} = \frac{4}{13}$$

b) It is a face card.

Face cards are: J, Q, K

Each suit has 3 face cards, and there are 4 suits.

Total face cards = $3 \times 4 = 12$

$$P(\text{Face card}) = \frac{12}{52} = \frac{3}{13}$$

4. Two cards are drawn from the pack of 52 cards. Find the probability that both are diamonds or both are kings.

Total no. of ways = ${}^{52}C_2$

Both are diamonds = ${}^{13}C_2$

Both are kings = ${}^{4}C_2$

P (both are diamonds or both are kings) = $({}^{13}C_2 + {}^{4}C_2) / {}^{52}C_2$

5. Two dice are tossed. The probability that the total score is a prime number is?

Clearly, n(S) = (6 x 6) = 36.

Let E = Event that the sum is a prime number.

Then E= {(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4,3),(5, 2), (5, 6), (6, 1), (6, 5)}

n(E) = 15.

P(E) = n(E)/n(S) = 15/36 = 5/12.

6. In how many different ways can the letters of the word 'VIDYAPEETH' be arranged in such a way that the vowels always come together?

VIDYAPEETH has 10 letters.

Vowels: I, A, E, E → 4 vowels (E repeats twice)
Consonants: V, D, Y, P, T, H → 6 consonants

Treat vowels as one group

Since vowels must always come together, treat them as one block.

So we arrange:

- 1 vowel block + 6 consonants = 7 items

Arrange these 7 items

All consonants are different.

Number of arrangements =

$$7! = 5040$$

Arrange vowels inside the block

Vowels: I, A, E, E

Arrangements of vowels:

$$\frac{4!}{2!} = \frac{24}{2} = 12$$

Total arrangements

$$7! \times \frac{4!}{2!}$$
$$5040 \times 12 = 60480$$

Answer: 60,480 ways

7. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is?

Let S be the sample space.
Then, n(S) = number of ways of drawing 3 balls out of 15 = $15C_3$ = $\frac{15*14*13}{3*2*1}$ = 455.
Let E = event of getting all the 3 red balls.
n(E) = $5C_3$ = $\frac{5*4}{2*1}$ = 10.
=> P(E) = n(E)/n(S) = 10/455 = 2/91.

MODULE III : (CLOCKS AND CALENDARS)

Short Answers

1. At what time between 5 to 6 O'clock will the hands of a clock point coincide with each other?

$$\theta = |30 \times H - \tfrac{11}{2} m|$$

$$\theta = |30 \times 5 - \tfrac{11}{2} m|$$

$$150 = \tfrac{11}{2} m$$

$$m = \frac{150 \times 2}{11} = \frac{300}{11} = 27\tfrac{3}{11} \text{ minutes}$$

$$\Rightarrow 5H\ 27\tfrac{3}{11} \text{ minutes}$$

2. What will be the angle between the hour and minute hands of the clock at 7:30?

7:30

$$\theta = |30 \times H - \tfrac{11}{2} m|$$

$$\theta = |30 \times 7 - \tfrac{11}{2} \times 30|$$

$$= |210 - 165|$$

$$= 45^\circ$$

3. If it is 5:40 in the 12 hours clock then what will be the time in the mirror?

To find mirror time in a 12-hour clock, use the rule:

$$\text{Mirror Time} = 11{:}60 - \text{Given Time}$$

Mirror Time = 11:60 - Actual Time

Given time 5: 40

Subtract from 11:60

Hours: $11 - 5 = 6$
Minutes: $60 - 40 = 20$

$$6:20$$

The mirror time of 5:40 is 6:20.

4. If the water image of the clock shows 3 hours 25 minutes, then what will be the actual time?

For a water image of a clock (reflection in water), the rule is:

$$\text{Actual Time} = 6:30 - \text{Water Image Time}$$

Actual Time = 6:30 - Water Image Time

Given water image time 3: 25

Subtract from 6:30

Minutes: $30 - 25 = 5$
Hours: $6 - 3 = 3$

$$3:05$$

The actual time is 3:05.

5. If 12th July 2010 is a Monday, what was the day of the week on 12th July 2009?

12 | July | 2009

$$\frac{\text{Date} + \text{month code} + \text{Century year code} + \text{Last two digits of the year} + \frac{L \cdot 2 \cdot 9}{4}}{7}$$

$$\frac{12 + 0 + 6 + 9 + 2}{7} = \frac{29}{7} = \text{Rem}^{t}\ 1$$

Sunday

6. The angle between the minute hand and the hour hand of a clock when the time is 4.20?

$$\theta = |30 \times H - \frac{11}{2} m|$$

4:20

$$\theta = |30 \times 4 - \frac{11}{2} \times 20|$$

$$= |120 - 110|$$

$$\boxed{\theta = 10^{\circ}}$$

7. At 3:40, the hour hand and the minute hand of a clock form an angle?

$$\theta = 130 \times H - \frac{11}{2} m$$

$$\theta = 130 \times 3 - \frac{11}{2} \times 360$$

$$= 190 - 220$$

$$= 110^\circ$$

$$\theta = 110^\circ$$

8. Today is Monday. After 61 days, it will be?

We need the day after 61 days from Monday.

Convert days into weeks

$$61 \div 7$$

61 = 8 \times 7 + 5

So, 61 days = 8 weeks + 5 days.

8 weeks = 56 days → same day (Monday)

Remaining 5 days

Count 5 days after Monday

1. Tuesday
2. Wednesday
3. Thursday
4. Friday
5. Saturday

Final Answer: Saturday.

Long Answers

1. A watch gains 4 seconds in 3 minutes and was set right at 8 AM. What time will it show at 10 PM on the same day?

A watch gains 4 seconds in 3 minutes.

Time from 8 AM to 10 PM

From 8:00 AM to 10:00 PM = 14 hours

Convert to minutes:

$$14 \times 60 = 840 \text{ minutes}$$

Gain in 840 minutes

The watch gains 4 seconds in 3 minutes.

$$\frac{840}{3} = 280$$

So, gain:

$$280 \times 4 = 1120 \text{ seconds}$$

Convert seconds to minutes:

$$1120 \div 60 = 18 \text{ minutes } 40 \text{ seconds}$$

Add gain to actual time

Actual time = 10:00 PM $10{:}00 + 18$ min 40 sec
= 10:18:40 PM

The watch will show 10:18:40 PM.

2. The clock was set on Thursday, at 4 AM. If the clock gains 20 minute per hour, then what will be the time that the clock shows on Friday, 8:30 PM?

Clock gains 20 minutes per hour.

Find actual time elapsed

From Thursday 4:00 AM → Friday 8:30 PM

Thursday 4 AM → Friday 4 AM = 24 hours

Friday 4 AM → Friday 8:30 PM = 16.5 hours

Total time:

$$24 + 16.5 = 40.5 \text{ hours}$$

Gain of the clock

Clock gains 20 minutes in 1 hour.

Total gain in 40.5 hours:

$$40.5 \times 20 = 810 \text{ minutes}$$
$$810 \div 60 = 13 \text{ hours } 30 \text{ minutes}$$

Add gain to actual time

Actual time = Friday 8:30 PM

Add 13 hours 30 minutes:

8:30 PM + 13 hours = 9:30 AM (Saturday)

30 minutes = 10:00 AM (Saturday)

The clock will show 10:00 AM on Saturday.

3. If March 1st, 2024 is a Saturday, what day of the week will be September 1st, 2024?

Given: March 1, 2024 = Saturday

Count days from March 1 to September 1

Days in the months:

March = 31, April = 30, May = 31, June = 30, July = 31, August = 31

Total days: $31 + 30 + 31 + 30 + 31 + 31 = 184$
Convert into weeks

184 = 26 \times 7 + 2

So, 184 days = 26 weeks + 2 days.

26 weeks → same day (Saturday)

Extra 2 days forward

Move 2 days after Saturday

Sunday (1), Monday (2)

September 1, 2024 will be Monday.

4. What was the day on 14 April 2000?

14 | April | 2000

$$\frac{14+0+6+00+0}{7} = \frac{20}{7} = \text{Rem "6"}$$

Friday.

5. What was the day on 16 August 1947?

16/ Aug/ 1947

$$\frac{16+3+0+47+11}{7} = \frac{77}{7} = \text{Rem } 0$$

Saturday

6. A watch which gains 5 seconds in 3 minutes was set right at 7 a.m. In the afternoon of the same day, when the watch indicated quarter past 4 o'clock, the true time is?

A watch gains 5 seconds in 3 minutes and was set correct at 7:00 AM.

The watch later shows 4:15 PM.

Time shown by the watch

From 7:00 AM → 4:15 PM

$$= 9 \text{ hours } 15 \text{ minutes}$$

Convert to minutes:

$$9 \times 60 + 15 = 555 \text{ minutes}$$

Gain of the watch

The watch gains 5 seconds in 3 minutes.

Number of 3-minute intervals:

$$\frac{555}{3} = 185$$

Total gain:

$$185 \times 5 = 925 \text{ seconds}$$

Convert to minutes:

$$925 = 15 \text{ minutes } 25 \text{ seconds}$$

Find true time

The watch is ahead by 15 min 25 sec.

$$4:15:00 - 15:25$$
$$= 3:59:35$$

The true time is 3:59:35 PM.

7. What was the day of the week on 11th March, 2026?

11 / march / 2026

$$\frac{11 + 4 + 6 + 26 + 6}{7}$$

$\frac{53}{7}$ Rem '4' wednesday.

8. On what dates of April, 2001 did Wednesday fall?

First find the day on 1 April 2001.

Step 1: Day on 1 Jan 2001

We know 1 Jan 2000 = Saturday.
Year 2000 is a leap year, so:

$$366 = 52 \times 7 + 2$$

So, 1 Jan 2001 = Monday.

Days up to April 1, 2001

January = 31 February = 28 March = 31

Total: $31 + 28 + 31 = 90$
90 = 12 \times 7 + 6

So, 6 odd days.

Starting from Monday:

Monday + 6 days = Sunday

So, 1 April 2001 = Sunday.

Find Wednesdays in April

If 1 April = Sunday, then:

3 April = Wednesday,10 April = Wednesday

17 April = Wednesday ,24 April = Wednesday

Wednesdays in April 2001 were on 3rd, 10th, 17th, and 24th.

MODULE IV: (DATA INTERPRETATION AND ITS APPLICATIONS)

SHORT ANSWERS

1.Study the following table and answer the questions.

Number of Candidates Appeared and Qualified in a Competitive Examination from Different States Over the Years

State	Year									
	1997		1998		1999		2000		2001	
	App.	Qual.	App.	Qual.	App.	Qual.	App.	Qual.	App.	Qual.
M	5200	720	8500	980	7400	850	6800	775	9500	1125
N	7500	840	9200	1050	8450	920	9200	980	8800	1020
P	6400	780	8800	1020	7800	890	8750	1010	9750	1250
Q	8100	950	9500	1240	8700	980	9700	1200	8950	995
R	7800	870	7600	940	9800	1350	7600	945	7990	885

A) Total number of candidates qualified from all the states together in 1997 is approximately what percentage of the total number of candidates qualified from all the states together in 1998?

$$\text{Required percentage} = \left[\frac{(720 + 840 + 780 + 950 + 870)}{(980 + 1050 + 1020 + 1240 + 940)} \times 100\right]\%$$

$$= \left[\frac{4160}{5230} \times 100\right]\%$$

$$= 79.54\% \approx 80\%.$$

B) What is the average candidates who appeared from State Q during the given years?

$$\text{Required average} = \frac{8100 + 9500 + 8700 + 9700 + 8950}{5}$$

$$= \frac{44950}{5}$$

$$= 8990.$$

2. The following pie-chart shows the sources of funds to be collected by the National Highways Authority of India (NHAI) for its Phase II projects. Study the pie-chart and answers the question that follow.

Sources of funds to be arranged by NHAI for Phase II projects (in crores Rs.)

A) Near about 20% of the funds are to be arranged through?

20% of the total funds to be arranged = Rs. (20% of 57600) crores

= Rs. 11520 crores

≈ Rs. 11486 crores.

Rs. 11486 crores is the amount of funds to be arranged through External Assistance.

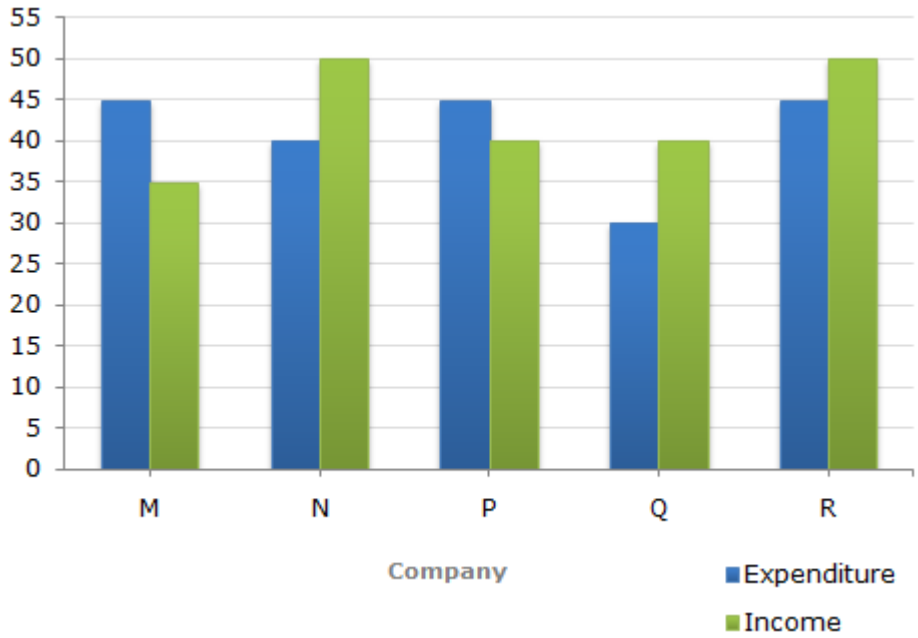
B) The central angle corresponding to Market Borrowing is?

Central angle corresponding to Market Borrowing = $\left(\frac{29952}{57600} \times 360°\right)$ = 187.2°

3. The following bar graph shows the Income and Expenditures (in million US $) of five companies in the year 2001. The percent profit or loss of a company is given by

% Profit/Loss =	**Income - Expenditure** **Expenditure**	**x 100**

Income and Expenditure (in million US $) of five companies in the year 2001.



A) The companies M and N together had a percentage of profit/loss of?

Total income of Companies M and N together

= (35 + 50) million US $

= 85 million US $

Total expenditure of Companies M and N together

= (45 + 40) million US $

= 85 million US $.

∴ Percent Profit/Loss of companies M and N together

$$\% \text{ Profit/Loss} = \left(\frac{85 - 85}{85} \times 100\right) = 0\%.$$

Thus, there was neither loss nor profit for companies M and N together.

B) In 2001, what was the approximate percentage of profit/loss of all the five Companies taken together?

Total income of all five companies

= (35 + 50 + 40 + 40 + 50) million US $

= 215 million US $.

Total expenditure of all five companies

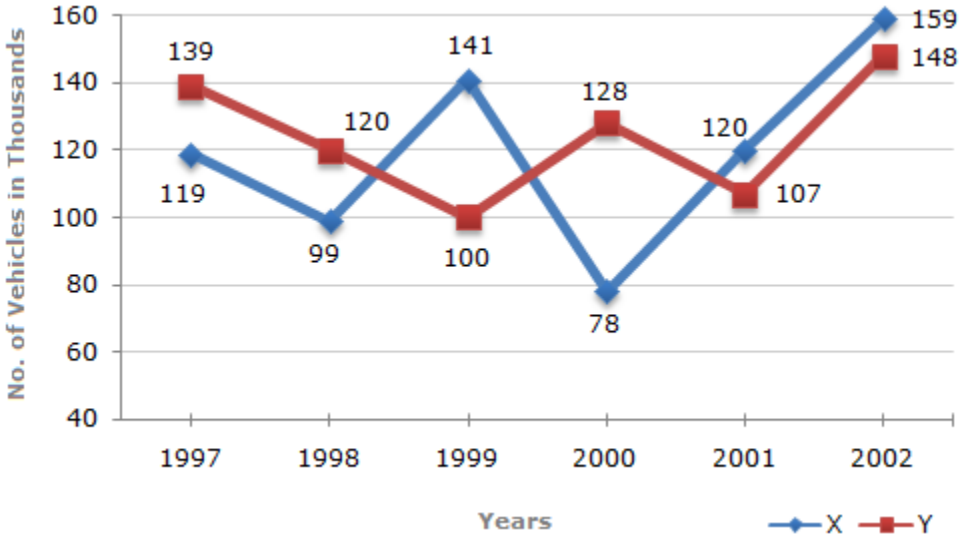
= (45 + 40 + 45 + 30 + 45) million US $

= 205 million US $.

$$\therefore \ \% \text{ Profit} = \left[\frac{(215 - 205)}{205} \times 100\right]\% = 4.88\% \approx 5\%.$$

4. Study the following line graph and answer the questions based on it.

Number of Vehicles Manufactured by Two companies over the Years (Number in Thousands)



A) What is the difference between the number of vehicles manufactured by Company Y in 2000 and 2001?

Required difference = (128000 - 107000) = 21000.

B) What is the difference between the total productions of the two Companies in the given years?

From the line-graph it is clear that the productions of Company X in the years 1997, 1998, 1999, 2000, 2001 and 2002 are 119000, 99000, 141000, 78000, 120000 and 159000 and those of Company Y are 139000, 120000,100000, 128000, 107000 and 148000 respectively.

Total production of Company X from 1997 to 2002

= 119000 + 99000 + 141000 + 78000 + 120000 + 159000

= 716000.

and total production of Company Y from 1997 to 2002

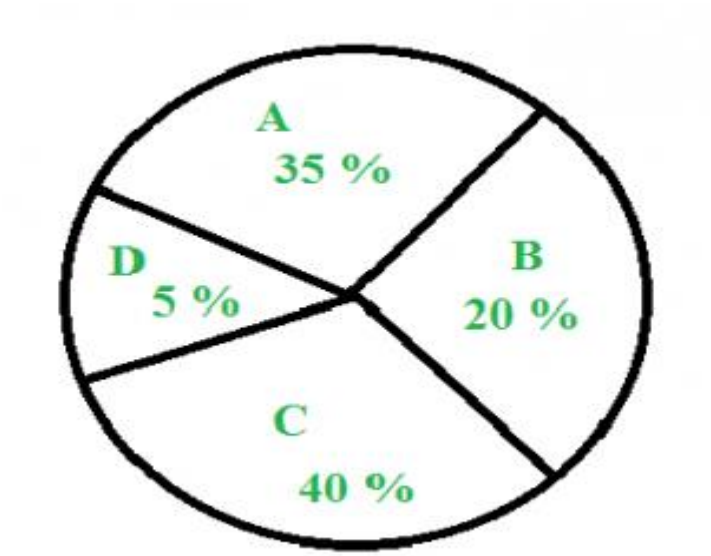
= 139000 + 120000 + 100000 + 128000 + 107000 + 148000

= 742000.

Difference = (742000 - 716000) = 26000.

Long Answers

1) Study the pie chart below and answer the questions that follow:



a) What is the central angle of type A?

Central angle of A = Percentage of A x 360 degrees = (35 / 100) x 360 = 126 degrees

b) If the total sale is 1200, what is the sale of B?

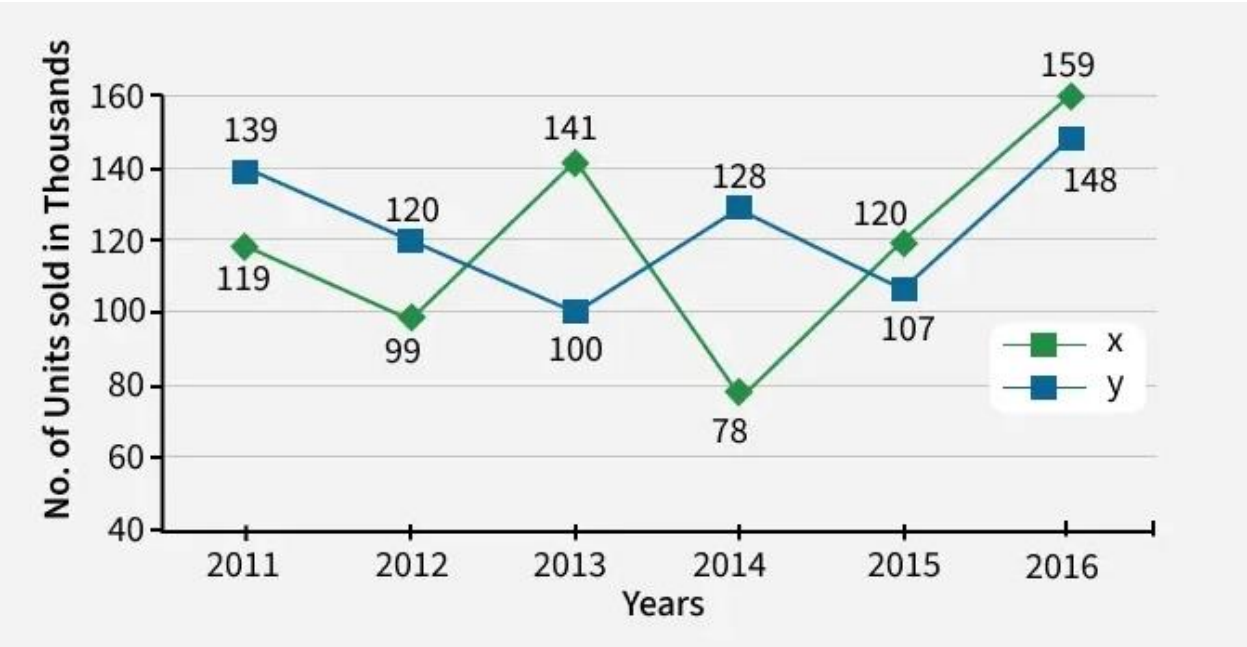
Sales of B = 20 % of 1200 = 240

c) What is the difference between the central angle of C and D?

Difference between the central angle of C and D = 40 % of 360 - 5 % of 360 = 35 % of 360 = 126 degrees

2) Study the following line graph and answer the questions based on it.

Number of Vehicles Manufactured by Two Companies over the Years (Number in Thousands)



a) What is the average number of vehicles manufactured by Company X over the given period?

Average number of vehicles manufactured by Company X

(119000 + 99000 + 141000 + 78000 + 120000 + 159000)/6 = 119333**.**

b) In which of the following years, the difference between the productions of Companies X and Y was the maximum among the given years?

The difference between the productions of Companies X and Y in various years are:

For 2011 (139000 - 119000) = 20000.

For 2012 (120000 - 99000) = 21000.

For 2013 (141000 - 100000) = 41000.

For 2014 (128000 - 78000) = 50000.

For 2015 (120000 - 107000) = 13000.

For 2016 (159000 - 148000) = 11000.

Clearly, maximum difference is 50,000 in 2014.

c) The production of Company Y in 2014 was approximately what percent of the production of Company X in the same year?

From the graph in 2014:

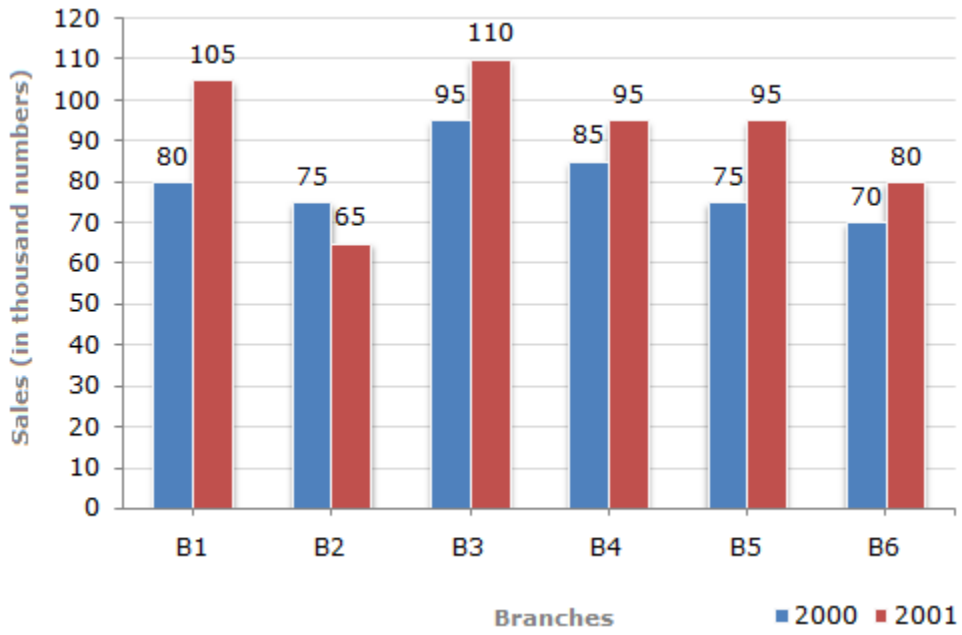
- Company X (green line) = 78 thousand = 78,000
- Company Y (blue line) = 128 thousand = 128,000

percentage = (128,000 / 78,000) × 100 ≈ 164.1 %

So, the production of Company Y in 2014 was approximately 164.1 % of the production of Company X's production in 2014.

3) The bar graph given below shows the sales of books (in thousand number) from six branches of a publishing company during two consecutive years 2000 and 2001.

Sales of Books (in thousand numbers) from Six Branches - B1, B2, B3, B4, B5 and B6 of a publishing Company in 2000 and 2001.



1.What is the ratio of the total sales of branch B2 for both years to the total sales of branch B4 for both years?

A) 2:3 B) 3:5 C) 4:5 D) 7:9

Required ratio = $\frac{(75 + 65)}{(85 + 95)} = \frac{140}{180} = \frac{7}{9}$.

2.Total sales of branch B6 for both the years is what percent of the total sales of branches B3 for both the years?

A)68.54% B)71.11% C)73.17% D)75.55%

Required percentage = $\left[\frac{(70 + 80)}{(95 + 110)} \times 100\right]\%$

$= \left[\frac{150}{205} \times 100\right]\%$

$= 73.17\%$.

3. What percent of the average sales of branches B1, B2 and B3 in 2001 is the average sales of branches B1, B3 and B6 in 2000?

A)75% B)77.5% C)82.5% D)87.5%

Average sales (in thousand number) of branches B1, B3 and B6 in 2000

$= \frac{1}{3} \times (80 + 95 + 70) = \left(\frac{245}{3}\right)$.

Average sales (in thousand number) of branches B1, B2 and B3 in 2001

$= \frac{1}{3} \times (105 + 65 + 110) = \left(\frac{280}{3}\right)$.

$\therefore$ Required percentage = $\left[\frac{245/3}{280/3} \times 100\right]\% = \left(\frac{245}{280} \times 100\right)\% = 87.5\%$.

4. What is the average sales of all the branches (in thousand numbers) for the year 2000?

A)73 B)80 C)83 D)88

Average sales of all the six branches (in thousand numbers) for the year 2000

$= \frac{1}{6} \times [80 + 75 + 95 + 85 + 75 + 70]$

$= 80.$

5.Total sales of branches B1, B3 and B5 together for both the years (in thousand numbers) is?

A)250 B)310 C)435 D)560

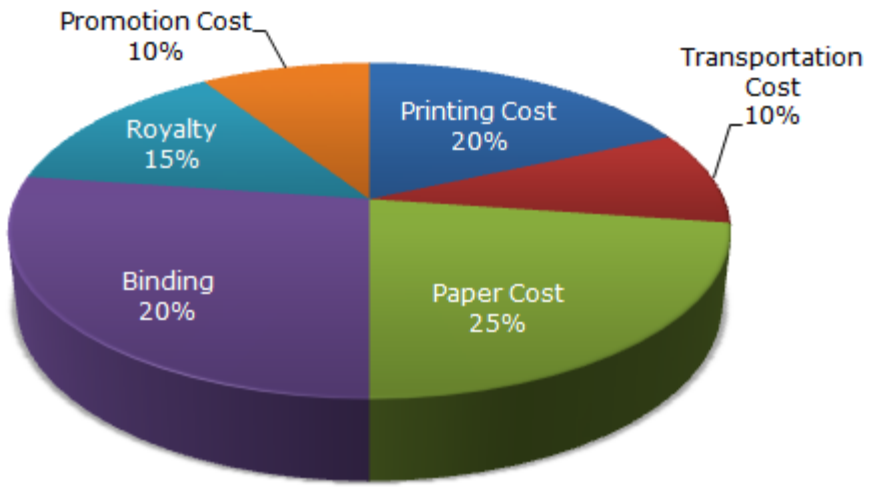
Total sales of branches B1, B3 and B5 for both the years (in thousand numbers)

$= (80 + 105) + (95 + 110) + (75 + 95)$

$= 560.$

4) The following pie-chart shows the percentage distribution of the expenditure incurred in publishing a book. Study the pie-chart and the answer the questions based on it.

Various Expenditures (in percentage) Incurred in Publishing a Book



1.If for a certain quantity of books, the publisher has to pay Rs. 30,600 as printing cost, then what will be amount of royalty to be paid for these books?

A.Rs. 19,450 **B.Rs. 21,200** **C.Rs. 22,950** **D.Rs. 26,150**

Let the amount of Royalty to be paid for these books be Rs. r.

Then, $20 : 15 = 30600 : r \quad \Rightarrow \quad r = \text{Rs.} \left(\frac{30600 \times 15}{20}\right) = \text{Rs. } 22{,}950.$

2.What is the central angle of the sector corresponding to the expenditure incurred on Royalty?

A.15° **B.24°** **C.54°** **D.48°**

Central angle corresponding to Royalty = (15% of 360)°

$$= \left(\frac{15}{100} \times 360\right)^{\circ}$$

$$= 54^{\circ}.$$

3.The price of the book is marked 20% above the C.P. If the marked price of the book is Rs. 180, then what is the cost of the paper used in a single copy of the book?

A.Rs. 36 B.Rs. 37.50 C.Rs. 42 D.Rs. 44.25

Clearly, marked price of the book = 120% of C.P.

Also, cost of paper = 25% of C.P

Let the cost of paper for a single book be Rs. n.

Then, $120 : 25 = 180 : n \Rightarrow n = \text{Rs.} \left(\frac{25 \times 180}{120}\right) = \text{Rs. } 37.50$.

4.If 5500 copies are published and the transportation cost on them amounts to Rs. 82500, then what should be the selling price of the book so that the publisher can earn a profit of 25%?

A.Rs. 187.50 B.Rs. 191.50 C.Rs. 175 D.Rs. 180

For the publisher to earn a profit of 25%, S.P. = 125% of C.P.

Also Transportation Cost = 10% of C.P.

Let the S.P. of 5500 books be Rs. x.

Then, $10 : 125 = 82500 : x \quad \Rightarrow \quad x = \text{Rs.} \left(\frac{125 \times 82500}{10}\right) = \text{Rs. } 1031250.$

$\therefore$ S.P. of one book = Rs. $\left(\frac{1031250}{5500}\right)$ = Rs. 187.50 .

5.Royalty on the book is less than the printing cost by:

A.5% B.33.5 C.20% D.25%

Printing Cost of book = 20% of C.P.

Royalty on book = 15% of C.P.

Difference = (20% of C.P.) - (15% of C.P) = 5% of C.P.

$\therefore$ Percentage difference $= \left(\frac{\text{Difference}}{\text{Printing Cost}} \times 100\right)\%$

$= \left(\frac{5\% \text{ of C.P.}}{\text{Printing Cost}} \times 100\right)\% = 25\%.$

MODULE V: (LOGICAL THINKING AND PATTERN ANALYSIS)

Short Answers

1. In a row of students, John is 8th from the left end, and Mary is 12th from the right end. Calculate the total number of students in the row.

John is 8th from the left, Mary is 12th from the right.

Total students = (John's position from left) + (Mary's position from right) – 1 = 8 + 12 – 1 = 19

Final Answer: The total number of students in the row = 19.

2. Radha selected the 27th card from the left in a row of 50 cards. What will be the position of the same card from the right?
Short Answer: Total cards = 50. Radha's card = 27th from left.
Position from right = 50 – 27 + 1 = 24.

Final Answer: The card is 24th from the right.

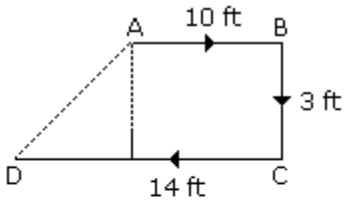
3. In a row of 54 persons, A is 35 from left side and B is 22 from right. Find the total number of persons between them.
A is 35th from left, B is 22nd from right → B's position from left = 54 – 22 + 1 = 33.
Difference = 35 – 33 = 2 → Persons between = 1.

Final Answer: There is 1 person between A and B

4. One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?

Udai facing = North

5. Reena walked from A to B in the East 10 feet. Then she turned to the right and walked 3 feet. Again she turned to the right and walked 14 feet. How far is she from A?



Required distance = AD

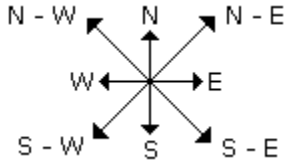
$$= \sqrt{3^2 + (14 - 10)^2}$$

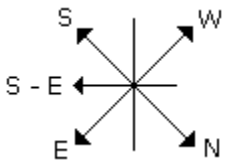
$$= \sqrt{9 + 16}$$

$$= 5 \text{ ft}$$

6. If SYMBOL is written as NZTMPC is it a certain code? How is NUMBER written in that code?

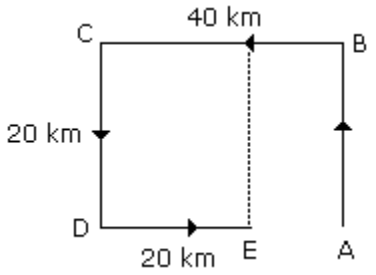
7. If South-East becomes North, North-East becomes West and so on. What will West become?

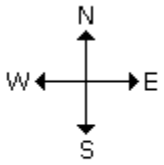




It is clear from the diagrams that new name of West will become South-East.

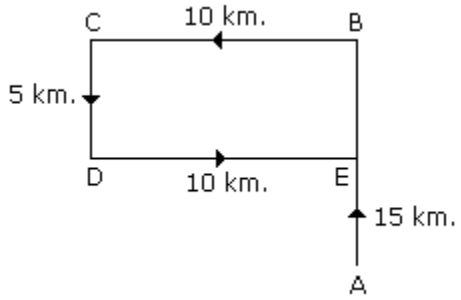
8. Sachin walks 20 km towards North. He turns left and walks 40 km. He again turns left and walks 20 km. Finally he moves 20 km after turning to the left. How far is he from his starting position?





Required distance = 40 - 20 = 20 km.

9. From his house, Lokesh went 15 km to the North. Then he turned west and covered 10 km. Then he turned south and covered 5 km. Finally turning to the east, he covered 10 km. In which direction is he from his house?



Therefore, it is clear that he is in the North from his house.

Long Answers

1. In a School, there are 147 people, the ratio of girls: boys is 1:6. Soumya is a girl who stands 15th from the top of that row and 7 girls are in front of her. How many boys are behind her?

Find number of girls and boys.

- Ratio = 1 : 6 → Total parts = 7.
- Girls = $\frac{1}{7} \times 147 = 21.$
- Boys = $\frac{6}{7} \times 147 = 126.$

Position of Soumya among girls.

- Soumya is 15th overall, with 7 girls in front.
- So, Soumya is the 8th girl.

Girls behind Soumya.

- Total girls = 21.
- Girls behind her = 21 – 8 = 13.

Boys in front of Soumya.

- Total persons before Soumya = 14.
- Girls before her = 7.
- Boys before her = 14 – 7 = 7.

Boys behind Soumya.

- Total boys = 126.

- Boys before her = 7.
- Boys behind her = 126 – 7 = 119.

Final Answer: The number of boys behind Soumya = 119

2. Sahil and Gaurav are standing in a row of persons. Sahil is 12th from left side and Gaurav is 18th from the right side of the row. If they interchanged their positions Sahil becomes 25th from left. What is the total number of persons standing in the row?

Ans: Initial positions.

- Sahil = 12th from left.
- Gaurav = 18th from right.

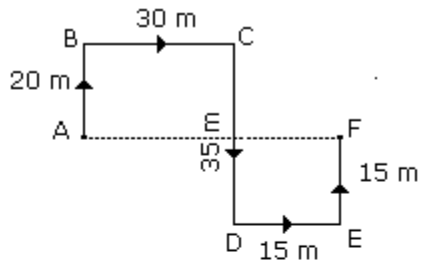
After interchange, Sahil takes Gaurav's position.

- Sahil becomes 25th from left.

Use formula: Total persons = Position from left + Position from right – 1. = 25 (Sahil's new position from left) + 18 (Gaurav's original position from right) – 1. = 42.

Final Answer: The total number of persons in the row = 42**.**

3. Rasik walked 20 m towards north. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many meters is he from the starting position?



Required distance = AF

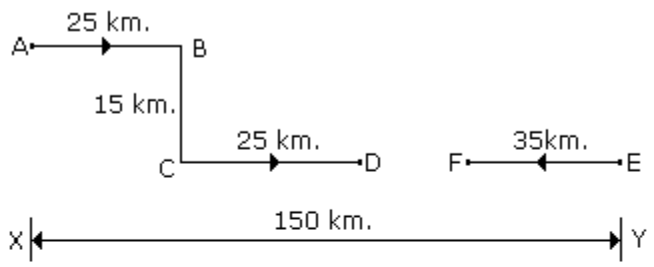
= 30 + 15

= 45 m.

From the above diagram, F is in East direction from A.

Hence the required answer is '45 m East'.

4. Two cars start from the opposite places of a main road, 150 km apart. First car runs for 25 km and takes a right turn and then runs 15 km. It then turns left and then runs for another 25 km and then takes the direction back to reach the main road. In the meantime, due to minor break down the other car has run only 35 km along the main road. What would be the distance between two cars at this point?



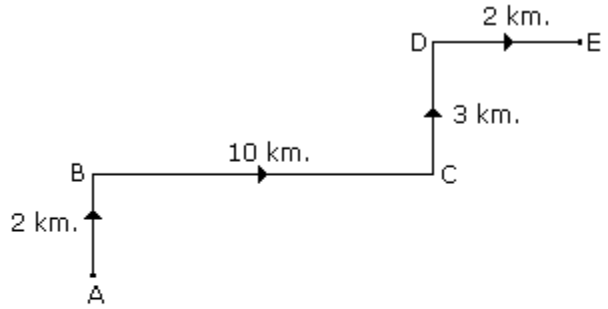
Required distance = DF

= 150 - (25 + 25 + 35)

= 150 - 85

= 65 km.

5. A man walks 2 km towards North. Then he turns to East and walks 10 km. After this he turns to North and walks 3 km. Again, he turns towards East and walks 2 km. How far is he from the starting point?



Required distance = AE

$= \sqrt{5^2 + 12^2}$

= 13 km.

6. In a certain code language, 'BLADE' is written as 'CKBCF' and 'CHALK' is written as 'DGBKL. How will DIGITAL' be written in that language?

Observe the coding pattern.

- BLADE → CKBCF Each letter is replaced by the next letter in the alphabet. (B→C, L→K, A→B, D→C, E→F).
- CHALK → DGBKL Again, each letter is shifted forward by +1. (C→D, H→G, A→B, L→K, K→L).

Apply the same rule to "DIGITAL".

- D → E
- I → J
- G → H
- I → J
- T → U
- A → B
- L → M

Combine the coded letters.
DIGITAL → EJHJUBM

7. In a certain code language, 'MAKE' is written as '32' and 'QUAD is written as '45'. How will 'SELF' be written as in that language?

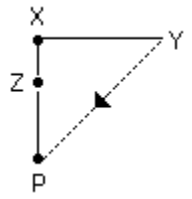
Find the sum of letter positions.

- M=13, A=1, K=11, E=5 → 13+1+11+5 = 30 → Code = 32 (i.e., +2).
- Q=17, U=21, A=1, D=4 → 17+21+1+4 = 43 → Code = 45 (i.e., +2).

Apply the same rule to "SELF".

- S=19, E=5, L=12, F=6 → 19+5+12+6 = 42.
- Add 2 → 42 + 2 = 44.

8. Y is in the East of X which is in the North of Z. If P is in the South of Z, then in which direction of Y, is P?



P is in South-West of Y.